

VariableTest Utility User Guide



Phoenix SecureCore Tiano™

Notices

Confidential and Proprietary Information

The contents of this document are confidential and proprietary to Phoenix Technologies Ltd. Access to this information is restricted. This document is provided for Distributor's internal use only. This document should not be disclosed to any third party, including customers.

Copyright

Copyright © 2010 Phoenix Technologies Ltd. All rights reserved. No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form or by any means, electronic, mechanical, magnetic, optical, chemical, manual, or otherwise, without the prior written permission of Phoenix Technologies Ltd.

Disclaimers

PHOENIX TECHNOLOGIES LTD. MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND WITH RESPECT TO THE INFORMATION HEREIN DESCRIBED AND SPECIFICALLY DISCLAIMS ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR ANY PARTICULAR PURPOSE OR NON-INFRINGEMENT. FURTHER, PHOENIX TECHNOLOGIES LTD. RESERVES THE RIGHT TO REVISE THIS DOCUMENTATION AND TO MAKE CHANGES FROM TIME TO TIME IN THE CONTENT WITHOUT OBLIGATION OF PHOENIX TECHNOLOGIES LTD. TO NOTIFY ANY PERSON OF SUCH REVISIONS OR CHANGES.

Trademarks

The following list contains the trademarks and/or registered trademarks of Phoenix Technologies Ltd. Exclusion from this list does not imply loss of trademark or registered trademark rights by Phoenix Technologies Ltd: Phoenix; TrustedCore; FirstBIOS; PhoenixBIOS; StrongROM; CoreArchitect; Phoenix Technologies; WinPhlash; WinFlash; Phoenix Technologies Logo; AwardBIOS; Embedded Simplicity; Embedded BIOS; StrongFrame; Firmbase; Purpose-Driven Firmware; and SecureCore Tiano.

Any use of other companies or legal entities' copyrights, trademarks, or registered trademarks is unintentional and in no way implies any legal claim to those copyrights, trademarks, or registered trademarks.

Document

The contents of this document are subject to change at the discretion of Phoenix Technologies.

Revision Date: March 22, 20110

TABLE OF CONTENTS

SCT VARIABLETEST UTILITY USER GUIDE	1
1 Introduction.....	1
2 Prerequisites.....	1
3 Launching the VariableTest Utility.....	1
4 Command Line Options.....	2

SCT VariableTest Utility User Guide

1 Introduction

The Phoenix SecureCore Tiano™ VariableTest utility is an EFI shell application that allows you to stress test the UEFI variable service by repeatedly performing READs/WRITEs to variables stored in NV storage.

Because the VariableTest tool runs in an EFI Shell environment, the BIOS must in turn support an EFI environment.

2 Prerequisites

The following prerequisite must be met before running VariableTest tool in an EFI shell.

- A USB key with or without EFI Shell boot loader. Copy the VariableTest tool execution file **VariableTest.efi** to the USB key.

Note: If you want to boot from the USB key, then you must copy the EFI Shell boot loader BootX64.efi to the **\efi\BOOT** directory of the USB key.

3 Launching the VariableTest Utility

Perform the following steps to start the VariableTest Utility.

1. Plug-in the USB key that you prepared in the “Prerequisites” section.
2. Activate EFI Shell at boot time.
 - a. If your USB key includes the EFI Shell boot loader, you can enter EFI Shell directly from this USB key at boot.
 - b. If your USB key does not include the EFI Shell boot loader, you can enter EFI Shell within the system BIOS by selecting **Internal Shell** from the Boot menu (**F5** key).
3. fs0 will display in the Device Mapping Table after booting to EFI Shell. Alternatively, you can use the "map -r" command to show the Device Mapping Table if you plugged in the USB key after booting to EFI Shell.
4. Type "**fs0:**" and press **Enter**. EFI Shell will change the working drive to fs0. The fs0 drive represents the USB key.
5. Type "**DIR**" and press **Enter** to view all the files in USB key. Verify that VariableTest.efi and the BIOS image exist in the list.
6. Run the VariableTest tool by typing the "**<command> [options]**" command.

4 Command Line Options

The following options can be used when launching the VariableTest utility.

Option	Description
<removed>	Get options from a file
<removed>	Show this help
<removed>	Do not display the VariableTest logo
<removed>	Detailed information (-vv=more)
<removed>	Turn off detailed process information
<removed>	Suppress all messages
<removed>	Turn on logging and specify the log file name. The default is : VariableTest.LOG
<removed>	Number of times to perform the test. The default is 1.
<removed>	Number of variables to test
<removed>	Size of each variable
<removed>	Generate variables
<removed>	Reset the system after test
<removed>	Clear generated variable after test
<removed>	List all variables created by this application